

Introduction to Data Anonymisation

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Radboud University 

You can find the latest version of this presentation at
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This powerpoint can be supplemented by material on the RDM website of the Radboud University, found at <https://www.ru.nl/rdm>

Topics covered in this presentation include:

- ❖ Why and when should you consider anonymisation?
- ❖ How does anonymisation fit in the overall security strategy of your research project?
- ❖ What is the level of anonymity that you should strive for?
- ❖ How to start planning your anonymisation strategy?
- ❖ What are the steps you can take to make anonymisation easier?
- ❖ Which steps and techniques can reduce the identifiability of research data?

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Who are we?

- ❖ **Research Data Management (RDM) Support Team**
- ❖ **The central hub for research data at the RU**
- ❖ **Located in the University Library**
- ❖ **Supported by datastewards in each institute**



Presenter name

We are the Research Data Management (RDM) Support Team. We function as the central hub for research data on the Radboud University. We assist researchers with everything related to research data, whether it is the research design, the collecting of data, the analysis of research data, the safe archiving or the publishing in line with the FAIR principles. We are based at the University Library and act as a central knowledge hub for the questions researchers may have with regards to any phase of the research cycle.

We are supported by datastewards in each institute. They have specialised knowledge of your field and are usually based within your department.

Why consider anonymisation?

As a researcher:



Easier sharing



More security



Fewer legal requirements

As a participant:



Sensitive topics



Protecting community



Easier participation

There are many good reasons why it can be beneficial to anonymise research data before making it public, both for the researcher and for the participant.

Researcher:

1. Anonymisation allows you to share or publish data that would otherwise be less suited for sharing or which would require more restrictions in order to be published or shared.
2. Anonymisation enhances the security of the personal data that you process, since there is no personal data in the dataset that could be leaked.
3. An anonymous dataset no longer has to adhere to data protection laws such as the GDPR, thus lowering the legal requirements you have to follow.

Participant:

1. Anonymisation lowers the risk for the participant when they are sharing sensitive, illegal or confidential information.
2. Anonymisation does not just protect the participant, but also people who they represent. It avoids stereotyping or stigmatisation of larger groups or locations.
3. Anonymisation might lower the barriers to participate in a study that would otherwise struggle to find participants.

Disclaimer

No one-size-fits all approach

Many variables

**Practices of discipline
Policies of research institutes
Intentions for sharing
Time, technology and money**

Good practices rather than guidelines

When in doubt, scream and shout



- ❖ There is no one-size-fits all approach for the anonymisation of research data.
- ❖ The implementation of anonymisation depends on the common practices of disciplines, the research data management policies of each institute, your intention to share the data and the time, money and technology available to anonymise the dataset.
- ❖ There are no all-encompassing guidelines, the best we can do today is discuss good practices.
- ❖ When in doubt, contact the authorities: datastewards

Table of content

1. Introduction

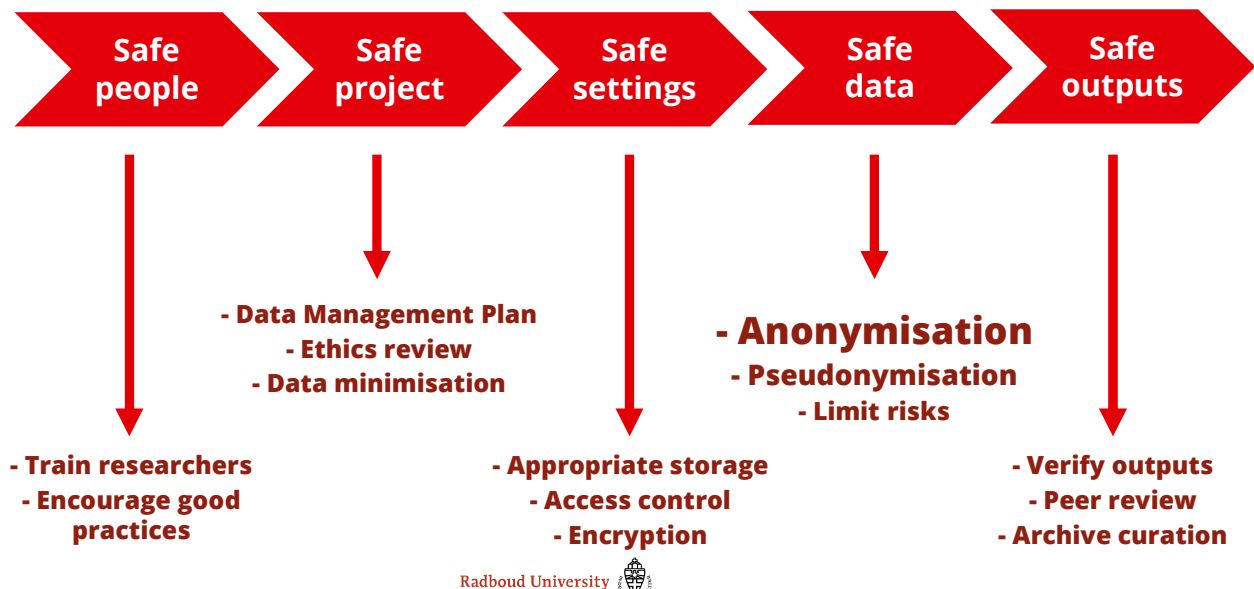
2. Anonymisation in theory

3. Anonymisation in practice

The presentation is broken down in three sections. The introduction will focus on the position of anonymisation in the overall security strategy for doing research. The second section will discuss some basic theoretical concepts related to anonymisation. The third section will provide some practical starting points by which it is possible to start thinking about anonymisation of research data.

INTRODUCTION

The 5 SAFEs



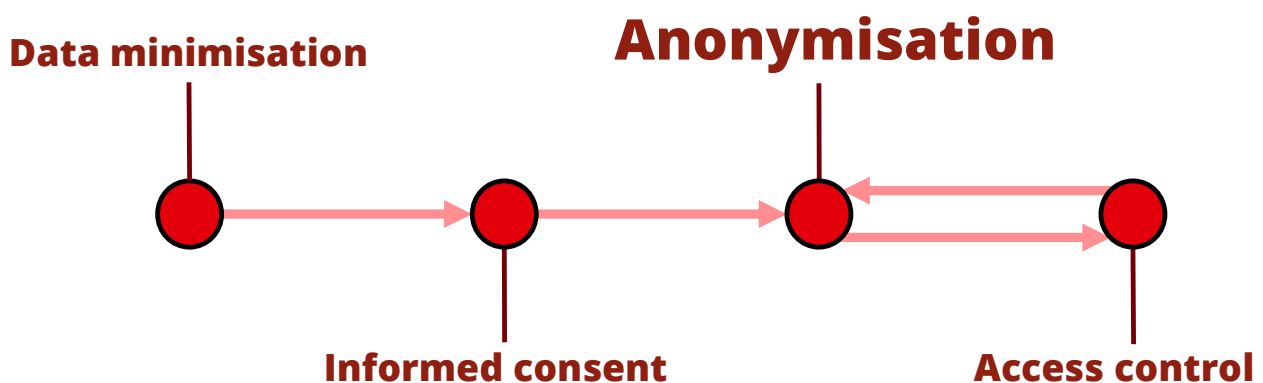
Anonymisation does not exist by itself, it is only one step in a much more elaborate chain of protective strategies that a researcher can employ to lower the risks of participants.

This security strategy is sometimes captured by the notion of the “5 SAFEs”

1. **Safe people:** make sure that researchers are capable to do their work safely (train researchers and encourage good research practices)
2. **Safe project:** take all necessary steps at the start of a project (create a data management plan, do an ethics review, don't collect more personal data than strictly necessary)
3. **Safe setting:** use the appropriate tools and technologies (appropriate storage, access control, encryption, etc.)
4. **Safe data:** take the necessary steps to reduce risks where possible (risk assessments, anonymisation or pseudonymisation)
5. **Safe outputs:** make sure that the output of research follow all protocols (peer review, archive curation, etc.)

The chain of protective strategies

Anonymisation is only one step in your overall security strategy



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Because anonymisation is only one step in a larger security strategy, you have to consider anonymisation in relation to three other security practices:

1. This starts with the concept of **data minimisation**, which means that the researcher will only collect the bare minimum of personal data that is required for a study.
2. Second a researcher can use the **informed consent** to inform participants about the personal data that they will collect, process and publish.
3. Third is the **anonymisation** of the dataset, which can remove the risks involved with personal data after it has been collected.
4. Lastly a researcher can choose **adequate access levels** in the repository where they archive or publish the data.

These last two steps are not definitive. As time moves on the security provided by anonymisation can lower. Technology may advance, new datasets may be linked together, causing the anonymisation to lose its value. In that case the security strategy must be reviewed in order to determine if past decisions are still adequate or must be updated.

Data minimisation

Make sure that you only collect the identifying data that is required for your study.



1. Preparation

- ❖ Intended research purpose
- ❖ Only required information
- ❖ Ethical committee



2. Process

- ❖ Existing sources of data
- ❖ Good survey questions
- ❖ Discuss with participants



3. Record

- ❖ At beginning of recording
- ❖ In a separate document
- ❖ Maximum privacy settings

It is easier to edit identifying data at the point of capture than it is to remove information after it has been recorded. This is why a key factor for the handling of personal data involves **minimizing the data that you process** to begin with. This means that personal data should only be processed with the intended (research) purpose in mind, and only to the extent that it is required for the study.

This can be done in various ways, for example by making a conscious selection in the questions that are included in the survey, by discussing what the participant wants to go “on record” during an interview or by tweaking the privacy settings of the software that is used. You can also look for existing archived sources of data that can be re-used.

If personal data must be collected it helps to do this in such a way that it can easily be identified and if necessary redacted. This can for example be done by capturing the identifying information within the first few minutes of the interview recording, so that it is easily edited out. It is also an option to transcribe personal data in a separate document, meaning that it can easily be removed from a participant’s file with the actual research data.

Informed consent

Explain:



What identifying data?



How will it be processed?



Can it be anonymised?

Be mindful:



Be explicit



Do not overpromise!



Discuss with participant

Ensure that participants are aware of your planned use of the data, and the (im)possibility that their information may be anonymised to protect them. Make it clear in your consent forms how extensively they will be de-identified (i.e., what elements will be replaced or removed). While direct identifiers may be eliminated (name, address, birthday, etc.), there may be other subtle clues to their identities that remain within the recording or transcript.

Agree in advance with participants which type of identifying information can be revealed in an interview. (For example, the participant may not wish to mention an employer's name). This is easier than removing the identifying data after the fact.

Keep in mind that not all data needs to be de-identified or anonymised. In some circumstances, you may be recording deeply personal accounts and should be mindful of a participant's right to have their story told in their own words. Some participants such as artists or politicians may also have a personal interest in staying identified.

Informed consent

How will the data be used?

I'm asking for permission to use anonymised quotations and narrative themes, along with any photographs and video you provide in the interviews or diaries for research purposes. All diaries and interviews from all participants will be analysed together for common themes about what everyday life is like when pregnant. As I work through my analysis, I will transcribe any audio recordings or handwritten diary entries. As I transcribe, I'll anonymise any identifying details, such as your name and address. All digital files will be saved on a password-protected computer at the Radboud University and all paper documents will be stored in a locked drawer at my office at the Radboud University, to which only I have access.

Throughout the project, I will be the only one with access to un-anonymised data, and my supervisors will have access to anonymised data. Since this project has gone through ethical approval, university staff may also audit this project to ensure I am protecting your information appropriately, and may ask to see sections of data.

I may need to break this confidentiality if you disclose illegal or criminal activity to me or I become aware of an issue that puts you or a child's safety at risk. In this instance, I will aim to first discuss the issue with you, but I may be legally obliged to share this information with the appropriate authorities.

This is an example of a section of the informed consent which discusses how anonymisation is handled. It is possible that the Ethics Committee of your research institute can provide similar templates which can be used for your study.

Anonymisation

“The principles of data protection should apply to **any information** concerning an **identified or identifiable natural person**.”

“To ascertain whether means are reasonably likely to be used to identify the natural person, **account should be taken of all objective factors**, such as the **costs** of and the **amount of time** required for identification, taking into consideration the **available technology** at the time of the processing and **technological developments**.”

“The principles of data protection should therefore not apply to anonymous information, namely **information which does not relate to an identified or identifiable natural person or to personal data rendered anonymous** in such a manner that the data subject is not or no longer identifiable.”



To understand what anonymisation is we have to look at the GDPR (Dutch: AVG) and how anonymity is handled there. In Recital 26 it is stated that the GDPR applies to “**any information** (meaning data and metadata, research and administrative data) concerning **identified** (meaning directly recognisable based on the data, e.g. names, photos) and **identifiable** (meaning traceable through combinations of data) **natural persons** (meaning living individuals, not businesses or organisations).

This means that the identifiability of a person in the dataset depends on a range of factors, of which the GDPR specifically mentions cost, time and current technologies. The GDPR also emphasises that these are not fixed in time, as technological development may change the likelihood that someone can be identified.

Ultimately an anonymous dataset has to either not contain any identifiable or identified information about a person, or contain personal data that has been made anonymous to a sufficient extent.

Effective anonymity

Full anonymity is an illusion:



The researcher knows



People close to participant may recognise them



You may need to archive raw data for integrity

The future is unpredictable:



Unexpected combinations



Future datasets



New technologies

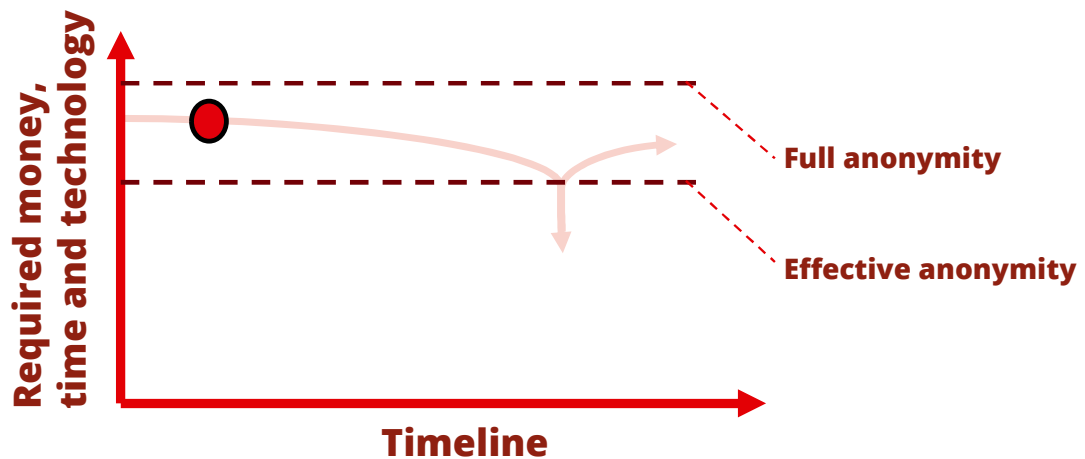
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The reality is that **full anonymity is in some ways an illusion**. There will always be people who can make a link between the “anonymous” data and the individuals who participated in a study. This is first and foremost the researcher who conducted the research, but also the people who are close to the participant and have more background information. Lastly there may also be explicit requirements that require you to securely archive the raw identifiable data so that scientific integrity can be established.

Another issue is that the **future will always be unpredictable**. A dataset that is seemingly anonymous today could lose this status in the future. This can happen because someone creates a combination of variables that you did not think of, because there could be future datasets that can be combined with your data or because future technologies make it significantly easier to identify anonymous participants.

The best you can strive for is therefore not “full anonymity”, but **effective anonymity**. This means that it would take an intruder a disproportionate amount of time, effort and expertise to identify a participant, or to reveal information about a participant that is not already publicly know.

Effective anonymity



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What is effective anonymity in your dataset is something that can be based on objective factors such as the **costs, time and the available technologies** that could be used to identify a person in your dataset. To be effectively anonymous it should take an intruder a disproportionate amount of time, effort and expertise to identify a participant. What this means in practice is something which differs for each dataset and can best be determined by the researcher and data steward.

However, it is also affected by the state of **technological development over time**. While the anonymisation today might be sufficient to stay above the threshold of effective anonymity, the future may bring new datasets, faster computers or systems of AI that could negate this anonymity. If technology develops to a point where effective anonymity is broken it is up to the researcher to review the new risks and take action where needed.

Anonymised text: [firstname1] [lastname1] is the wife of [occupation1] [lastname2]. [pronoun] is a mother of [numeric] children; [numeric] boys and a girl. [firstname1] is educated to university level and that is where [pronoun] met [pronoun] future husband. [firstname1] dresses elegantly and is often seen carrying out charity work. However, [pronoun] is a mum first and foremost and the interactions we see with [pronoun] children are adorable. [firstname1]'s sister, [firstname2], has followed [firstname1] into the public eye. [pronoun] was born in [date1] and will soon turn [numeric]. When pregnant, [firstname1] suffers from a debilitating illness called [otherattribute1], which was little known about until it was reported that [firstname1] had it.

This is an example to show how anonymity can degrade with time. The above text could be considered anonymous based on the information which has been removed. However, the limited information which is in there is still sufficient to identify the person.

You

which celebrity was educated at the university, had a famous sister and suffered from a debilitating illness during pregnancy

ChatGPT

Another celebrity who fits this description is Catherine, Duchess of Cambridge, formerly known as Kate Middleton. She studied at the University of St. Andrews and is well-known for being the sister of Pippa Middleton.

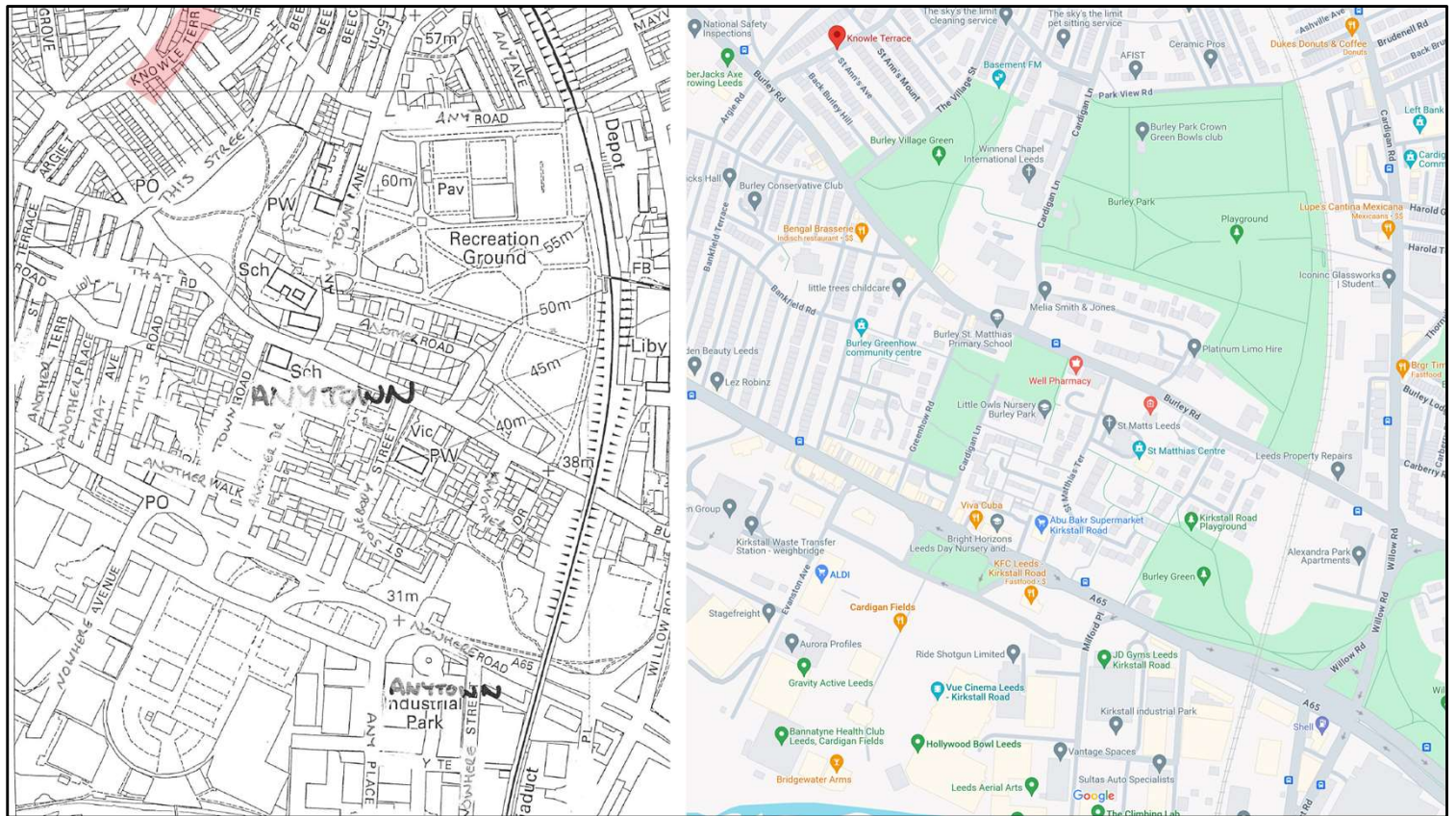
During her pregnancies with her children, Kate Middleton experienced a severe form of morning sickness called hyperemesis gravidarum. This condition causes extreme nausea, vomiting, and dehydration, leading to significant health challenges during pregnancy. Kate's struggles with hyperemesis gravidarum were widely reported in the media, and she has been open about her experiences, raising awareness about this condition that affects some pregnant women.

To retrieve the seemingly anonymous information of the previous slide we simply have to ask a service such as ChatGPT. This makes it exceptionally easy to find out exactly who the person in question would be.

Anonymised text: [firstname1] [lastname1] is the wife of [occupation1] [lastname2]. [pronoun] is a mother of [numeric] children; [numeric] boys and a girl. [firstname1] is **educated to university level** and that is where [pronoun] met [pronoun] future husband. [firstname1] dresses elegantly and is often seen carrying out charity work. However, [pronoun] is a mum first and foremost and the interactions we see with [pronoun] children are adorable. **[firstname1]'s sister, [firstname2], has followed [firstname1] into the public eye.** [pronoun] was born in [date1] and will soon turn [numeric]. When pregnant, [firstname1] **suffers from a debilitating illness called [otherattribute1],** which was little known about until it was reported that [firstname1] had it.

Original text: Kate Middleton is the wife of Prince William. She is a mother of 3 children; 2 boys and a girl. Kate is educated to university level and that is where she met her future husband. Kate dresses elegantly and is often seen carrying out charity work. However, she is a mum first and foremost and the interactions we see with her children are adorable. Kate's sister, Pippa, has followed Kate into the public eye. She was born in 1982 and will soon turn 40. When pregnant, Kate suffers from a debilitating illness called Hyperemesis Gravidarum, which was little known about until it was reported that Kate had it.

Upon testing it turns out the strategy was right, the information which is missing can easily be added based on the retrieved information.



This is another example to prove the same point. The map on the left is seemingly anonymous, as almost all street names and locations have been edited out or have been made nameless. Yet due to the availability of Google Maps one minor detail on the edge of the map (Knowle Terrace) is sufficient to identify the city in an instant (Leeds, UK).

Access Controls

The level of anonymity in your dataset also determines the access level of publishing.

Open Access	Registered Access	Restricted Access
Anonymous data	Potentially identifiable data Pseudonymised data	Sensitive data Directly identifiable data
Access without a login	Access via a login	Access with explicit approval
License as agreement	Data use agreement	Signed data use agreement

Access control can also be achieved via embargoes, which delay the release of data.

There is a clear difference between releasing data to the world at large and making it available to a smaller, more defined group. With public release, it may be virtually impossible for you to retract the data, for example if there is identifiable information included that should not have been released. You also do not have control over the actions and intentions of the recipients of that information. This is why **an appropriate access level is often the most important step in your security strategy.**

The level of anonymity in your dataset determines the access level of publishing. Most repositories provide two or more access levels that determine how others can access the research data, and consequently how easy it is to trace who has access to the data.

This slide provides an example that is based on the Radboud Data Repository, the institutional repository of the Radboud University. The first is **open access**, which is meant for anonymous data which can be accessed by anyone and is only regulated by means of a license. At the other end there exists **restricted access**, which is meant for directly identifiable or sensitive research data. Access to this data requires a signed data use agreement and explicit approval of the researcher, making it relatively easy to trace who has access to the data. The Radboud Data Repository offers registered access, which is meant for pseudonymous research data. This does not require explicit permission of the researcher but re-users still sign a simpler data use agreement, meaning that they can be traced.

An embargo can also provide a solution in some cases, allowing data to remain secure for a certain period of time before it gets released to the public.

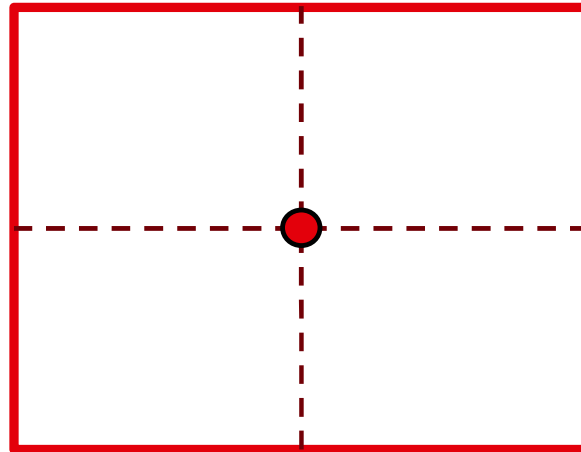
Balancing protective strategies

1. Data minimisation

3. Anonymisation

2. Informed consent

4. Access control



In conclusion, the researcher needs to pick a strategy that is suitable for their study.

1. Sometimes this might entail minimizing the identifiable data that is collected at the start of a study. However in other cases it might be unavoidable to capture certain personal information.
2. In that case the informed consent might make it possible to work with these sensitive data with permission of the participant.
3. Depending on what is agreed in the informed consent anonymisation might be the key to lower the sensitivity after the data is collected.
4. When this is too costly, time-intensive or comes at the expense of the information value then access control (either restrictions or embargoes) are the measure of last resort, allowing researchers to work with identifiable data regardless.

The researcher thus has to locate where on the above square they want to locate their research project. This can be different for every study and can be determined together with the data steward.

ANONYMISATION IN THEORY

Choose your strategy



Respecting participants



Assessing the risks for participants



Assessing the information loss

A good starting point for your anonymisation is to ask yourself several questions at the start of your research, all of which are born from the notion that the **researcher should respect the rights and interests of their participants**. In more sensitive cases it can also help to discuss these topics with your participants.

Secondly, it is good to assess the **actual risks for the participants**. Not all data leaks are equally harmful for your participants, and not every dataset might be at equal risk of outside intrusion. Moreover, different intruders might have different goals when they try to access a certain dataset without authorisation. This is called **adversary modelling** and determines how much anonymisation is required to protect your participants.

Thirdly, you have to ask yourself **how much information is lost** when you decide to anonymise the dataset. If you anonymise every identifiable detail you might lose too much relevant research information, while not anonymizing anything at all leaves your participants at risk. The **anonymisation spectrum** helps you to find a balance between these two.

Respecting participants

Anonymisation can have major ethical complications



Minorities can become invisible or underrepresented



Names or locations have social, cultural or historical meaning



Anonymity can dampen the societal impact or delay change



Participants may want to be public and receive recognition

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Anonymisation is not just a story about security of the data or the protection of your participants, there can also be complex ethical considerations. There exists a tension between how you can keep your participants safe and yet respect their personal interests. This is best discussed with the ethical committee of your research institute.

One problem can be that minorities become invisible or less represented when they are anonymised due to their “outlier” status. While being an identifiable group presents a risk, consistently anonymizing them also “erases” them from sight.

Some names and locations also have important social or cultural meanings which are erased when they are anonymised, negating the cultural meaning or history of the people or locations that you study.

By keeping participants anonymous you could also dampen the societal impact of your study or delay essential change for participants in a vulnerable position. Your study might identify a problem but due to the anonymity of participants policy makes will be unable to respond.

Lastly you can also have participants who explicitly ask to be identifiable for the recognition this brings them. Examples of this can be artists, politicians or other people who might benefit from the exposure.

Adversary modelling

Ask yourself what could go wrong

Name	Problem	How
Singling out	Identifying an individual in a dataset	Unique values Unique combinations of values
Linkability	Combining multiple records to identify someone	Multiple values in a dataset Combinations between datasets
Inference	Deducing information about someone with a level of certainty	Correlations in the dataset Profiles of participants

Research does this all the time!



A good starting point for your anonymisation strategy is to ask yourself the question **what could possibly go wrong** with the data that you collect in your study.

Some obvious problems that could be considered data leaks are:

1. **Singling out:** this occurs when it is possible to identify an individual in the dataset, for example because of unique (combinations of) values or simply because there are names or other direct identifiers present in the dataset.
2. **Linkability:** this occurs when it is possible to combine multiple records to identify someone, for example because there are multiple values in a dataset or because different datasets can be linked to each other.
3. **Inference:** this occurs when the dataset makes it possible to deduce information about someone with a level of certainty, through correlations between the data or because it is possible to create profiles of participants.

These three “risks” are problematic for researchers, because the very act of doing research involves making links between datasets or inferences between different variables. Research is thus actively creating these risks in a controlled manner in order to obtain relevant and meaningful results.

Adversary modelling

Ask yourself which “adversary” might be interested in the data



A second question that is relevant to adversary modeling is to ask yourself **who the potential adversary for the data might be**. The goal should always be to keep the risks for the participants reduced to a minimum, but what risks there are depends mostly on the contents of the dataset, but also on the type of intruders (adversaries) that might be interested in identifying the person behind the data. One relevant question can therefore be to ask yourself what type of person might try to de-identify your dataset, what risks does this present for your participants and what means do they have available to them to do this de-identification?

1. A **marketeer** will generally have limited means to de-identify a dataset of potential clients, since he or she has the aim to contact as many individuals as possible. They would only be successful if a large fraction of the dataset is re-identified and have limited means at their disposal to achieve this goal. This means that they generally represent a lower risk than other adversaries.
2. A **journalist** might have more means to identify participants and presents more risk for participants, since they seek to target a specific individual but might not know if information of that participant is contained in the dataset. They might also have more means to identify participants.
3. A **prosecutor** would have extensive means to identify someone and introduces major risks to your participants, because it can be assumed that they already know that data about the individual is contained in the dataset.

Depending on who your “adversary” is you will have to take different measures, for which anonymisation can be one possible solution.

Anonymisation spectrum

There is a balance between anonymity, data utility and information loss

Raw identifiable (personal) data

Anonymised information



Based on the risks that exist for the participants you must decide **up to what level you want to anonymise** the dataset. This is a balance that is found somewhere on the spectrum between identifiable (personal) data and totally anonymised information.

Anonymisation always comes at a price, since you might also lose information that you might want to keep for research purposes. Ultimately this information loss can also reduce the data utility of the dataset. This means that you cannot just assume that anonymisation is always the best option, and that you must make a balanced decision with regards to the extent to which you can anonymise a dataset.

You should **avoid over-anonymising**, since removing too much information can distort data, making it unusable, unreliable or misleading. It is generally better to control access than to over-anonymise.

NSC PRIMARY INTEREST

THE WHITE HOUSE
WASHINGTON

LBJ LIBRARY
Mandatory Review
Case # NLJ 93-46876
Document # 5

September 23, 1964

~~TOP SECRET~~

MEMORANDUM FOR THE PRESIDENT

Re: Summary of the existing plans for emergency use of nuclear weapons

On March 26 you approved recommendations from McNamara and the Joint Chiefs to put into effect updated instructions for expenditure of nuclear weapons in emergency conditions.

This instruction covers four emergency situations. Two of them are

[REDACTED]

In these two cases the commanders

[REDACTED]

The other two cases are

[REDACTED]

The instructions reveal an interesting difference between situations in which nuclear weapons would do enormous civilian and industrial damage and situations in which they would be used in the upper atmosphere or on the high seas.

[REDACTED]

It is possible that we ought to take account of this distinction in anything we say in the next few days.

McG. B.

PARTIALLY DECLASSIFIED
ON 09/23/93 BY 468
NARA Doc. 12-111196

~~TOP SECRET~~

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4. Cohen's Illegal Campaign Contributions

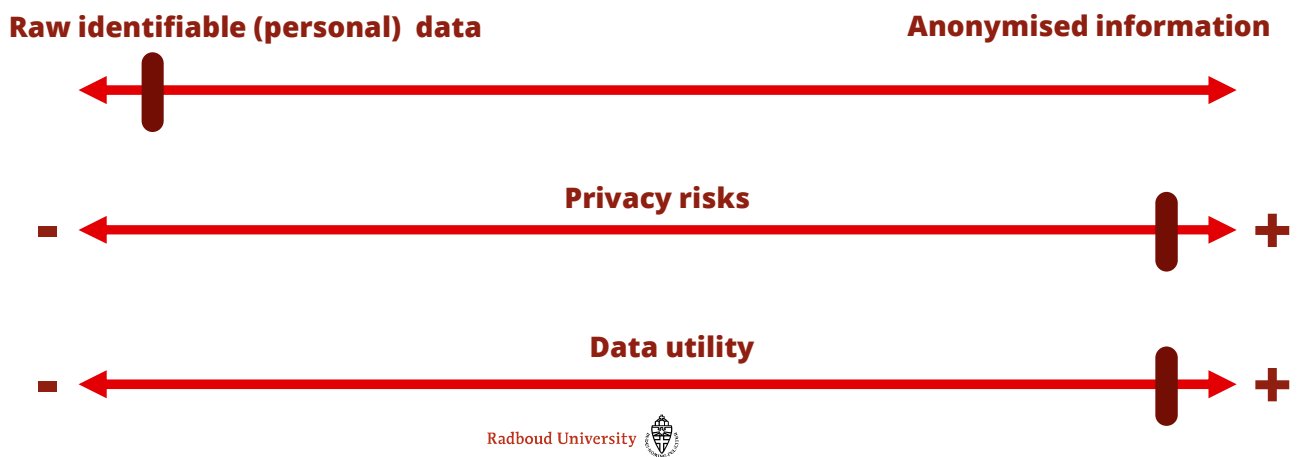
On approximately June 16, 2015, Individual-1, for whom Cohen worked at the time, began an ultimately successful campaign for President of the United States. Cohen had no formal title with the campaign, but had a campaign email address, and, at various times advised the campaign, including on matters of interest to the press. Cohen also made media appearances as a surrogate and supporter of Individual-1. (PSR ¶ 39).

During the campaign, Cohen played a central role in two similar schemes to purchase the rights to stories – each from women who claimed to have had an affair with Individual-1 – so as to suppress the stories and thereby prevent them from influencing the election. With respect to both payments, Cohen acted with the intent to influence the 2016 presidential election. Cohen coordinated his actions with one or more members of the campaign, including through meetings and phone calls, about the fact, nature, and timing of the payments. (PSR ¶ 51). In particular, and as Cohen himself has now admitted, with respect to both payments, he acted in coordination with and at the direction of Individual-1. (PSR ¶¶ 41, 45). As a result of Cohen's actions, neither woman spoke to the press prior to the election. (PSR ¶ 51).

These are two examples of what can go wrong when anonymisation is not applied correctly. The first example is likely "overanonymised", as so much information has been redacted that the information has become altogether useless. The second example shows a situation in which there has likely been too little anonymisation, since the context makes it all too easy to identify who Individual-1 would be.

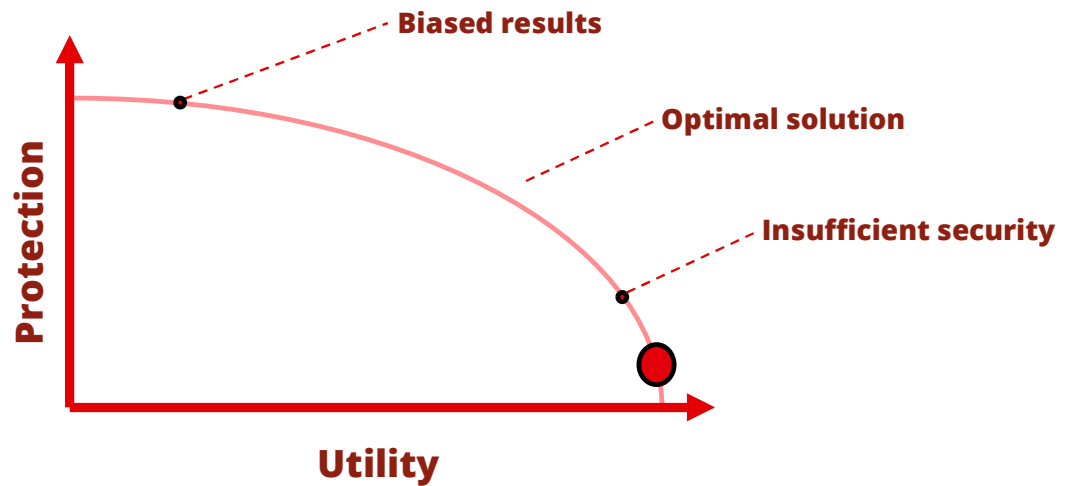
Anonymisation spectrum

There is a balance between anonymity, privacy risks and data utility



When you start to anonymise identifiable research data you are reducing the privacy risks for your participants. However, this comes at a price, since you are also losing information that you might want to keep for research purposes. Ultimately information loss can also reduce the data utility of the dataset.

Anonymisation spectrum



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Ideally the anonymisation that you apply to the research data exists somewhere on the spectrum between protection and utility. The data must be sufficiently secure to avoid data leaks or other problems, but not so strongly over-anonymised that the dataset contains biased, misleading or useless research data.

ANONYMISATION IN PRACTICE

Where to begin?



Start early!



Locate time & budget



Informed consent



Agree as a team



Ask for help



Be involved

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A good start to any anonymisation effort would be to **start early** in your research. Planning during the design phase what you will do with the personal data in your research is a lot easier than producing solutions when you already have collected the data. You can for example design a survey with closed questions rather than open questions. When transcribing audio data you can also plan ahead and already indicate identifying information with a searchable symbol, for example #. This will make it easier to look for elements of the data that you need to anonymise.

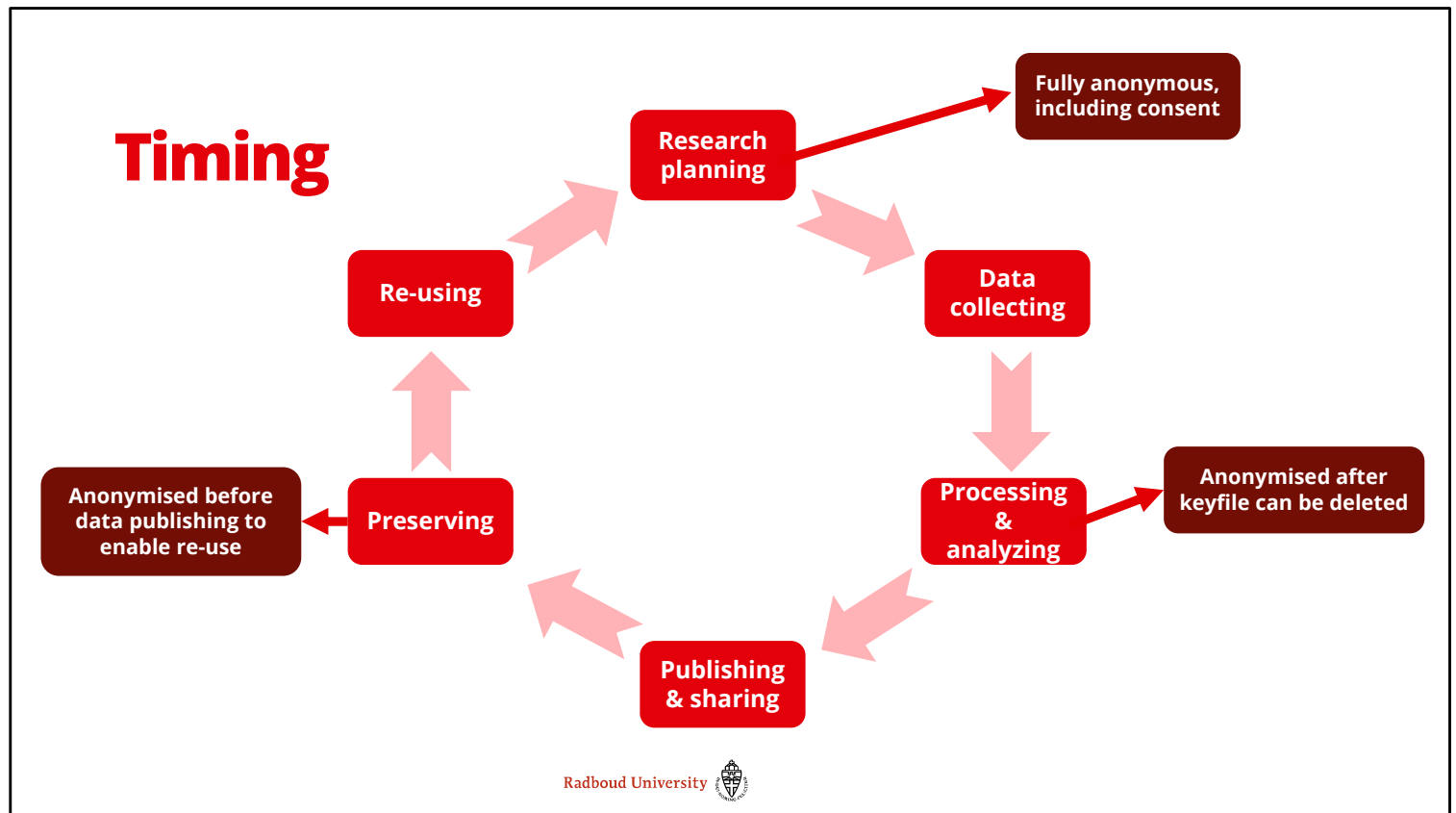
You should also factor the time and expense of anonymisation into the funding of your research project, allowing you to take care of this process without practical constraints.

Anonymisation is something which you should explain in the informed consent. Try to accurately describe what you will do with the personal data that people give to you. Don't make promises that you cannot keep (for example full anonymity) and be realistic about any risks that might exist for your participants.

This also applies to research that involves multiple people, it is important that the entire team agrees on a strategy and sticks to this to avoid mistakes or accidental disclosures.

You are not alone in this process. Many repositories have expertise with anonymising datasets and can help you if you upload your data to their service. The datasteward can also advice and assist within your research institute.

Asking for help is great, but don't outsource the anonymisation process altogether. It is important that you stay involved yourself, because nobody knows the data like you do. External parties do not know what the sensitive parts of the data there are or could miss details that are evident to you.



Depending on the research design the anonymisation might occur at the different times during the research process.

In some cases research can occur completely anonymous, meaning that no identifiable information is collected and every step, including informed consent, occurs anonymous.

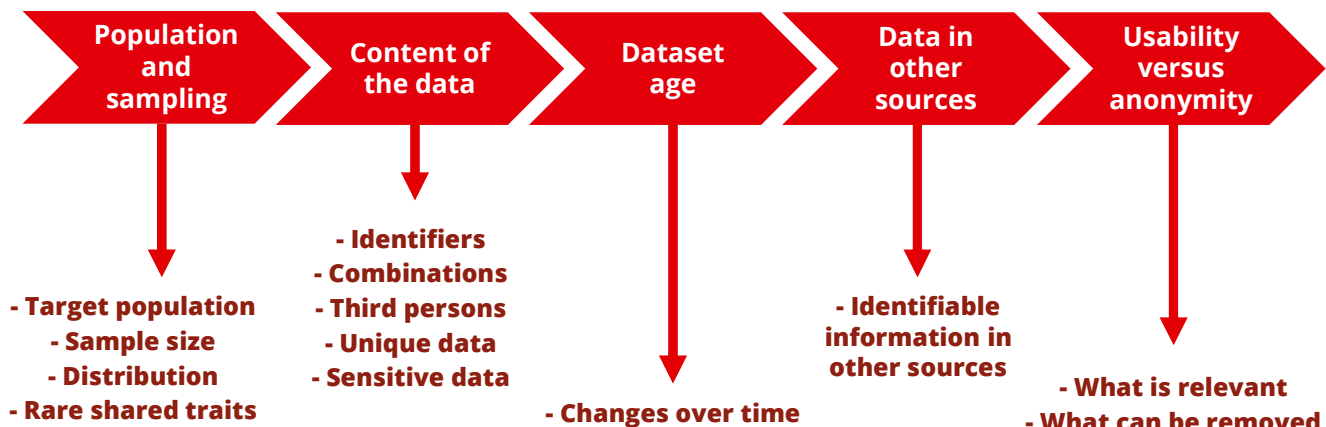
More commonly it is necessary to collect some identifiable information, either because this is necessary for the research data, or because there is contact information of participants involved. In that case anonymisation begins after the data is collected, but not right away. Ethical requirements, such as the necessity to report accidental findings or the right to be forgotten that your participants means you can only start the anonymisation process after the time specified in the informed consent has expired.

A third option would be to anonymise the research data at the end of the research process, when the data has to be archived and could be useful for other researchers in anonymised form. Here the goal is to make the data more accesible.

Anonymisation plan

[Click here!](#)

- ❖ **More detailed than a data management plan**
- ❖ **A record of decisions and their argumentation**
- ❖ **Both planning in advance and documentation afterwards**



It is good to plan ahead when it comes to anonymisation. The first step for this can be filling out a data management plan, which covers a broad spectrum of topics related to data management, including anonymisation and pseudonymisation.

Sometimes it can however also be useful to make a more detailed plan specifically aimed at the anonymisation process, in which you systematically assess all the risks of the dataset and the steps you intent to take to reduce or remove these. The anonymisation plan records both the decisions and the argumentation why these decisions were taken. You can find a template for such an anonymisation plan at <https://doi.org/10.5281/zenodo.10782781>.

A good anonymisation plan will look at various topics, including:

1. **Population and sampling:** who is the target population, how big is the sample, what are the common or rare traits of the population?
2. **Content of the data:** which identifiers are included in the dataset, is any data exceptional or sensitive?
3. **Dataset age:** have the data of the population changed over time?
4. **Other sources:** is it possible to connect other sources, or can individuals be identified based on other sources?
5. **Usability versus anonymity:** what is the significant, essential or relevant information that must be preserved? What can be removed?

Besides a way of anticipating risks and preparing for the research project the anonymisation plan can also serve as useful documentation for the archiving of the dataset, since it provides a good overview of the identifiers that you have located and the anonymisation steps that you decided to take.

Keeping track of your work



Keep original data as long as needed



Work consistently



Use clear indicators



Clean up syntax, meta-data & documentation



Keep a record of the changes



Erase original data when appropriate

Radboud University 

Besides an anonymisation plan that can be made during the planning, you can also document the anonymisation process during the research project.

In general it is good to **keep the original data** for as long as needed. What this means depends on institutional policy. For integrity reasons you often still need to preserve the raw, anonymised data for at least 10 years, regardless of the anonymisation process.

Whenever you are anonymizing it is recommended that you use **clear indicators**, such as [brackets].

While you work you can **keep a record of the anonymisation**. The level at which this is necessary depends on the level of detail you want to document.

It is critical that you **work consistently**, meaning that you apply anonymisation throughout the research project, for all participants and all variables. If you fail to anonymise a single case of identifiable information it might jeopardise the entire dataset.

For good anonymity is also important that you **clean up other locations** where there might be identifying information present, such as the syntax, metadata or documentation.

In some cases it might be possible or even necessary to **erase the original data**. Depending on the type of data this might require certain procedures to avoid copies being left behind.

Interview	Page	Original data	Changed to
1	1	Age 27	Age range 20-30
1	2	Spain	European country
1	2	20 th June	June
1	3	Amy	Moirra
1	3	Manchester	English provincial city
2	2	Aunty Jean	Aunt
2	2	Francis	My friend
2	3	Dunberry School	A primary school
2	3	Supermarket manager	Senior executive
2	4	North Colchester	North part of town

This is an example of a very detailed anonymisation logbook. It can be used during the research process, but is at the same time also a risk factor. After all, this log defeats the entire process of anonymisation. This means that care must be taken to delete the logbook before the data is published or shared.

Changes made on 16-02-2024 by Henry van der Burgt:

Identifier	Changed to	Example before	Example after
Names	Replacement	Jennifer	[Sofia]
Age	Ranges of 10	34	[30 – 40]
City name	National level	London	[British city]
Specific dates	Removed day	13-05-2011	[May 2011]
Relations	Removed name	Uncle Charles	[uncle]
City locations	Removed name	Luna Pharmacy	[local pharmacy]
Job name	Generalised	CEO	[senior executive]
Income	Generalised	€ 8600 a month	[high income]

This is a less detailed anonymisation log that still provides insight in the anonymisation process without creating any risks for the participants. The examples are not from the dataset but fictional. Due to the vagueness of this log this is information which you can include in the documentation of the research project. It is also something which can be used to develop institutional standards for anonymisation.

Blanket anonymisation

One bad solution is to “blindly” delete or pseudonymise all names, locations and other identifiers in a dataset.



Loss of information value and context



There are no clear boundaries



Very difficult with large datasets



Impossible with mixed methods

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A seemingly easy method for anonymisation is to blindly “blanket” anonymise all possible identifiers in a dataset. While this might seem practical, it is rarely the best approach and often highly problematic.

- 1) This type of anonymisation comes at the expense of a lot of information value and meaningful context of the data.
- 2) It is unclear where this approach begins and ends. How to handle family members, third parties, people mentioned in passing? How to maintain the information of social networks?
- 3) This is particularly difficult when it comes to anonymizing large datasets, since this is costly implement in a way that is still somewhat useful for research purposes.
- 4) With mixed methods research different methods tend to counteract the anonymisation of other parts of the study, making this very difficult to anonymise. While you could for example anonymise the survey-part of the study, this would be negated by the interviews with the same participants.

Because of these limitations it is almost always better to make a considered decision what will be anonymised and what will be kept.

Differences in variables

Identifier type	Type of data
Direct identifier	Personal data directly relatable to an individual
Indirect (quasi) identifier	Information that may uniquely identify an individual (when linked with other identifiers or datasets)
Sensitive identifier	Information that is subject to legal or ethical concerns

- ❖ **Sometimes it is not clear (e.g. medical conditions)**
- ❖ **Both direct and quasi identifiers can be sensitive.**
- ❖ **Not all sensitive data is an identifier.**

A better approach to anonymisation classifies identifying information in several types of identifiers.

1. **Direct identifiers** are personal data that is directly relatable to an individual, for example name, address, postcode, telephone number, picture, e-mailaddress, keyfile, BSN number, IP address, social media or a geographical location with less than 20.000 people. But also names of family members, unique information such as hospital IDs, clinical trial record numbers, account numbers exact dates, photo or video recordings and biometric or human genome identifiers.
2. **Indirect or quasi-identifiers** refer to information that may uniquely identify an individual, for example when linked with other identifiers or datasets. This can for example be workplace, occupation, education, salary, age, gender, region, income. But also certain geographic identifiers, ethnicity, immigration status, membership of an organisation, household composition, number of children, general dates.
3. A separate class of identifiers concerns **sensitive identifiers**, which are subject to legal or ethical concerns and thus require extra attention. This could be criminal history, sexual preferences and behaviour, political affiliations, medical records and so on.

While this seems a straightforward distinction, in practice it is rarely completely evident.

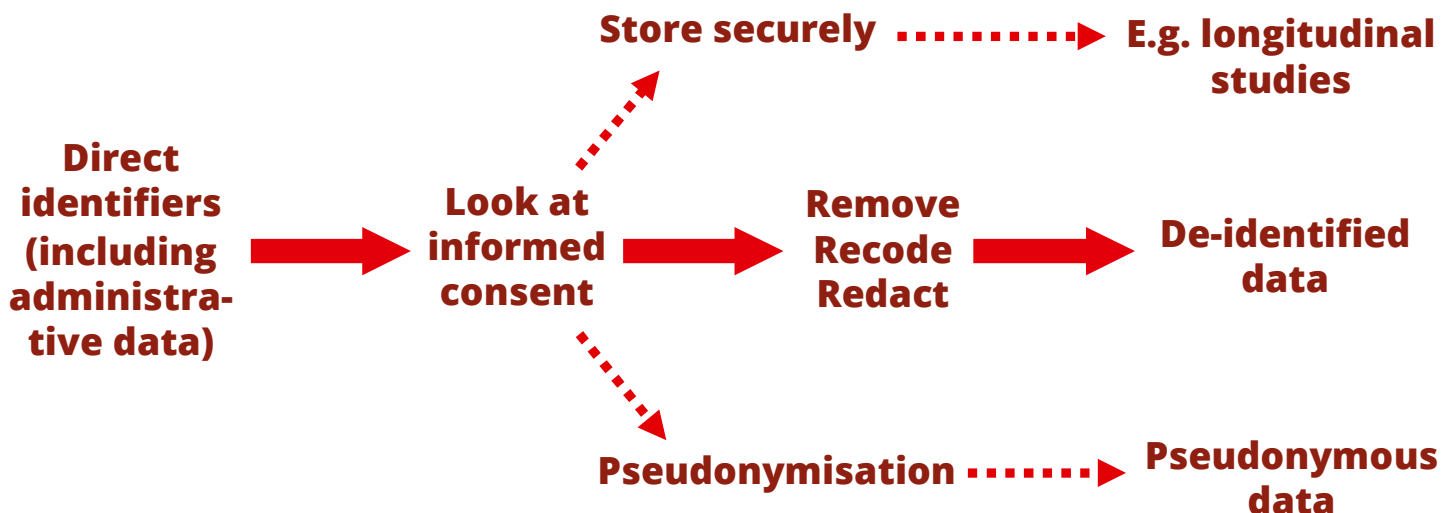
1. Medical information can be directly identifying when it is a very rare medical condition or a quasi-identifier when the medical issue is more common. The sensitivity of medical data also depends on a range of factors.
2. The distinction between direct and quasi-identifiers exists separately from the sensitivity of the data. Data can be both a direct or indirect identifier and at the same time sensitive.
3. Just because data is sensitive in nature does not automatically make it a direct or quasi identifier. There can be anonymous data that is still sensitive, for example because it reflects on a group as a whole.

Direct or indirect?

Direct identifier		Indirect identifier
	Joe Biden j.biden@prez.us Male 81 years old Pennsylvania Avenue 1600 Washington DC \$400,000 per year 	

This is a short exercise by which people can sort direct and indirect identifiers.

Step 1: Direct identifiers



The first step to any anonymisation process is to look up the direct identifiers in the dataset and to decide based on the informed consent what must be done with them. In most cases this means that they will be redacted, recoded or deleted in order to effectively remove them from the dataset. The end result of this step is sometimes called de-identified data, which is not the same as anonymous data.

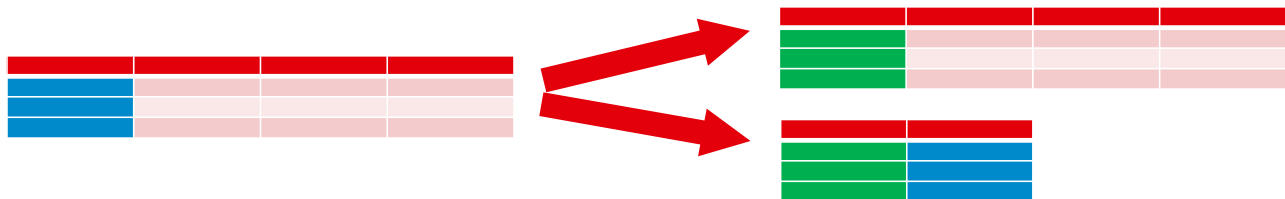
Alternatively, it is possible to store this data securely, for example when the research project involves longitudinal data gathering for which people must be contacted in the future.

A third option is to pseudonymise the direct identifiers. This is not a true anonymisation technique but can be an alternative when anonymisation is excessively difficult, time-consuming or would come at the expense of too much information value.

Be mindful of the so-called “privacy paradox”, meaning that you do not have to retain the keyfile of a study if its only role is to enable participants to exercise their privacy rights. When all the data in the study is anonymous retaining the keyfile actually creates a privacy risk, meaning that it is often better to delete the keyfile. After deletion of the keyfile the dataset will be fully anonymous and thus no longer subject to the requirements of the GDPR, meaning people do not have the right (or possibility) to exercise their rights anymore.

Pseudonymisation

Techniques that replace, remove or transform information that identifies individuals, and keep that information separate.



Benefits	Downsides
Possible when anonymisation is not	Privacy risks not entirely removed
It is possible to verify data accuracy	Requires periodic evaluation of safety
Reduces risks for data breach	Requires safe storage of keyfile

Pseudonymisation involves techniques that replace, remove or transform information that identifies individuals, and keep that information separate. Because of this it is not a true anonymisation technique, data remains identifiable as long as the separate information exists. Because of this the data must also be treated as personal data under the GDPR.

Compared to the raw data pseudonymisation does offer some advantages. It is easier to implement than anonymisation, it leaves a link by which to verify the accuracy of the data and it effectively reduces the risks for a data breach. Downsides are also evident however, because there remains a link between the participant and the research data the privacy risk is not entirely removed and remains a vulnerability that must be securely stored and processed.

Pseudonymisation vs anonymisation

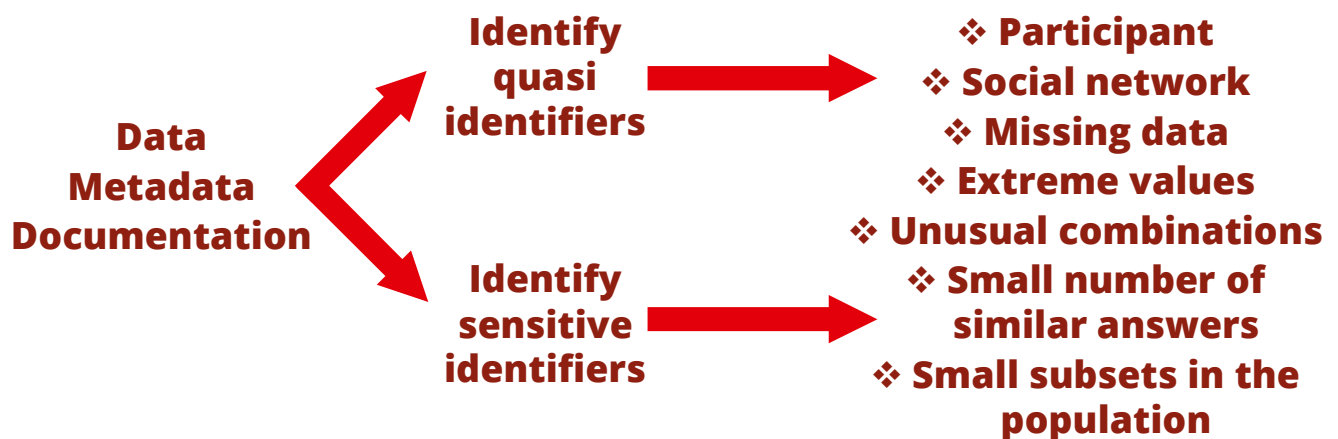
Technique	Identifiers	Reversible?	Legal status
Anonymisation	Direct identifiers deleted	Irreversible	No personal data
Pseudonymisation	Direct identifiers separated	Reversible	Still personal data

Variable	Raw data	Variable	Pseudonymous	Variable	Anonymous
Patient	91542	Participant	4105		
Name	Bob Burns				
City	Nijmegen	Region	Gelderland	Country	Netherlands
Date of birth	15-08-1962	Year of birth	1962	Age	60-70
Income	8,453	Income	7,500-10,000	Income	5,000 – 15,000
Job	Judge	Job	Legal	Job	Legal
Car	Ferrari	Car	Ferrari	Car	Sportscar
		Keyfile	4105 = Bob Burns		

The key difference between anonymisation and pseudonymisation is the link between participant and research data, often in the shape of **direct identifiers**. With anonymous data these have been deleted, making the restoration of this connection between participant and research data impossible and thus effectively removing the personal data from the dataset. With pseudonymous data this link is only separated, making the connection still reversible and the data effectively still personal data.

In the examples of this slide it is possible to see the differences. Raw data contains all the personal data, including direct identifiers name and patient number. Pseudonymous data already lowers the exactness of many of these indirect identifiers, while it replaces the directly identifiers with a participant number, which can still be linked via the keyfile. Meanwhile anonymous data removes the link created with the keyfile, while it lowers the identifiability of the indirect variables even further.

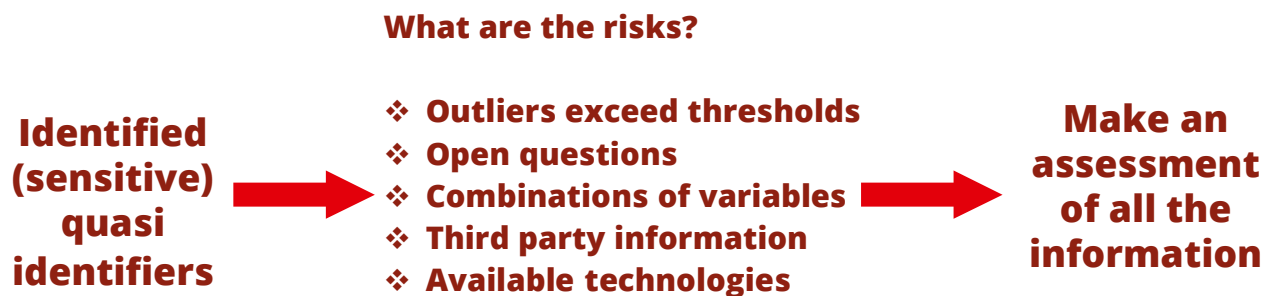
Step 2: Identify quasi / sensitive identifiers



Step two focuses on the quasi-identifiers and sensitive identifiers. These are often less obvious in the dataset than the direct identifiers. Identifying them in the data will generally take more time and effort. A full search requires that you look at a variety of different areas where there can be potential quasi or sensitive identifiers. It is important that you look at both the data, metadata, documentation and other data included in the study, because any of these could contain quasi or sensitive details.

When looking for quasi or sensitive identifiers you can focus on the characteristics of the participant themselves, but also on the people in their social network who are present in your dataset. But it is also smart to look at missing data and extreme values in the data. Thirdly it is useful to screen for unusual combinations of quasi-identifiers or a small number of similar answers. Lastly it might be helpful to screen for small subsets in the population which might be relatively easy to identify.

Step 3: Identify risky quasi identifiers



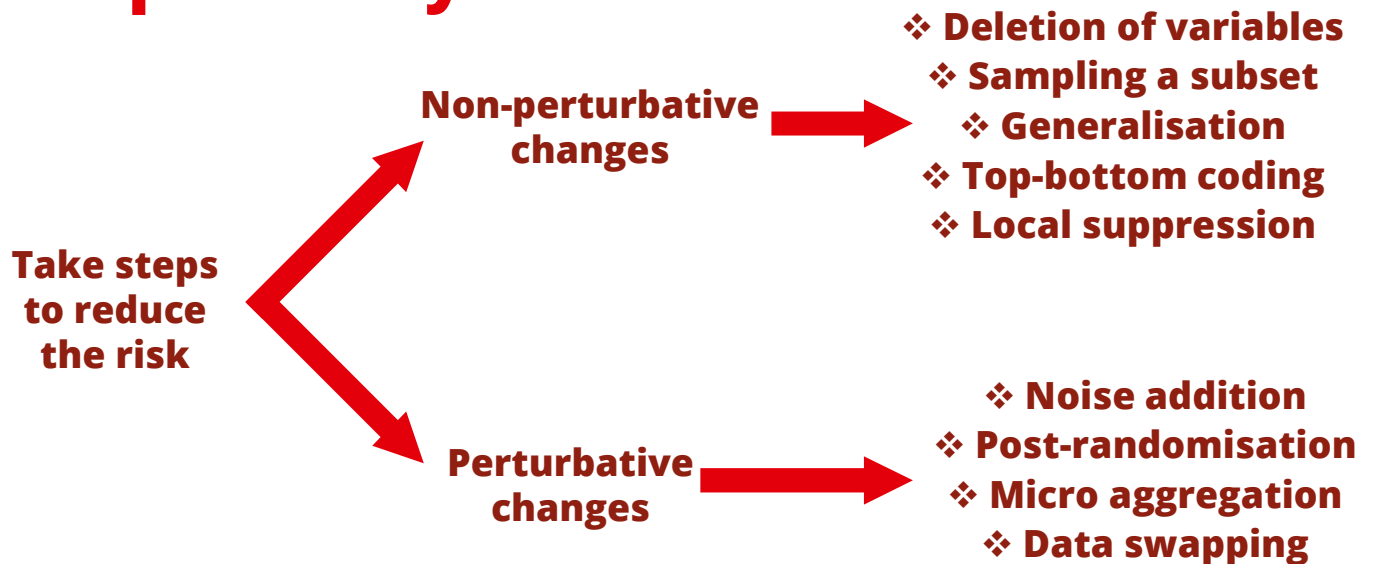
After step 2 you will likely have identified many quasi-identifiers and sensitive identifiers in the dataset. Not all of these are equally “risky” however. Some might need careful consideration because they do pose a risk to the participants, while others might be essentially risk-free. This means you must assess each of the quasi and sensitive identifiers for a variety of risk factors.

This can be because extreme values in the dataset exceed certain thresholds. Another reason could be because it is an open question that entices risky answers that must be screened individually. There can also be combinations of certain variables that increase the risks for participants. Another risk to pay attention to is the third-party information that is already (publicly) available and that could be matched with your dataset. Lastly this is also something that depends on the available technologies. Just because data is relatively safe today does not mean that it will remain so in the future.

One way to assess the risky quasi-identifiers is to describe an individual to a friend using only the information available in the dataset. Is there any likelihood that the person would be recognizable? In other words, what is the chance someone is recognizable, and at what level, if you take all (contextual) information into account?

Ultimately the goal is to assess these findings so that you know what the risks are and which variables you must address in the next step. This not only makes you aware of the risks of your dataset, but also can serve as a paper trail that shows due diligence if any data breaches or accidents where to occur.

Step 4: Anonymise!



In step 4 you can actually start to anonymise the dataset. There are many ways in which you can go about this. Some of them are fairly straightforward and easy to implement, others can be complex requiring you to be familiar with statistical analysis and specialist software. For this workshop we limit ourselves to some of the simpler and practical ones.

Broadly speaking there are two types of changes that you can make to risky (sensitive) quasi-identifiers in order to make them less identifiable. There are **non-perturbative changes** that remove certain information, but leave the overall dataset intact, and **perturbative changes** that actually change the structure of the dataset. Which of these techniques are the best option in your study is really something which can only be established on a case-by-case basis, often together with the data steward.

Deletion (redaction) of variables

Consider dropping risky identifiers completely

Benefits	Downsides
Definitive removal of sensitive identifiers	Potentially high information loss
Easy when variables are irrelevant	Distortion of statistical properties
Last resort when other options fail	Limits possibilities for re-use

Sometimes a simple solution can be to delete a variable in its entirety, assuming that it is not relevant to answer the research question.

This has some obvious benefits; the removal is definitive and thus a good measure of last resort if no other anonymisation techniques will help. However it is also an extreme solution, coming at the expense of high information loss, potential distortion of statistical properties and limited possibilities for re-use.

Generalisation

Generalise data to reduce the precision of the information

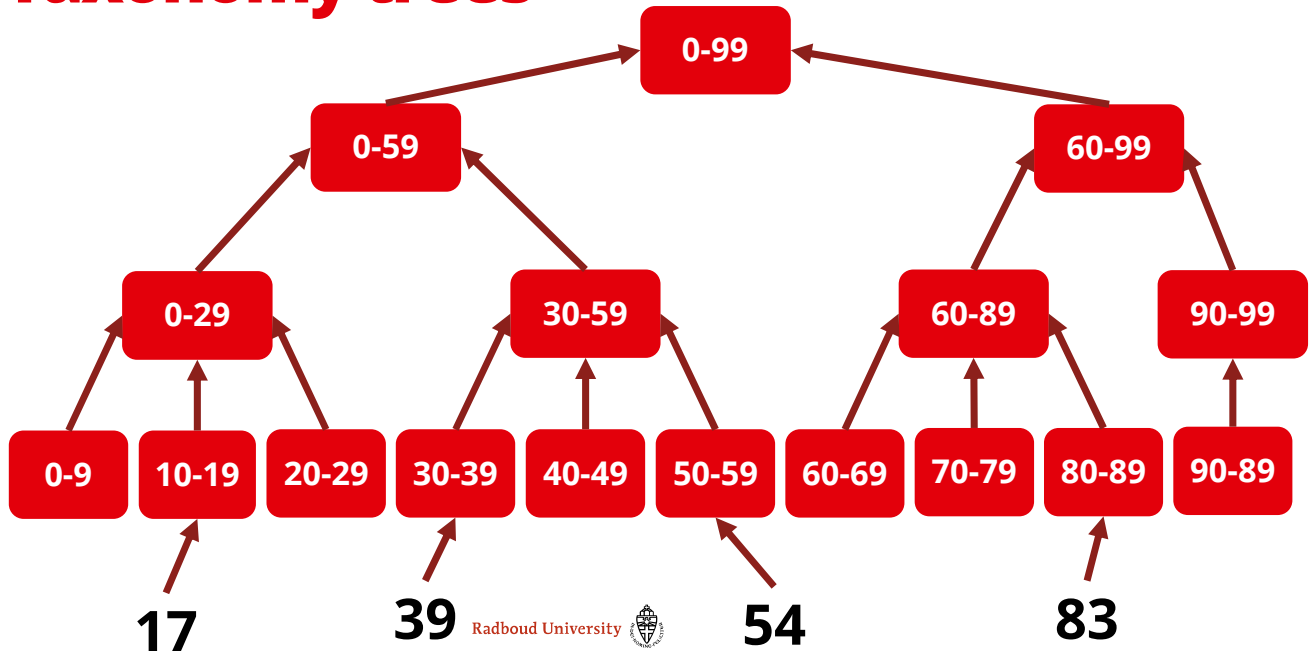
72 years → 70 to 80 years 6525 XZ → 6524 - 6526
Amsterdam Osdorp → Amsterdam € 1400 → Low income

Benefits	Downsides
The scale of this change is relative	Not possible with two answer options
Easy to do in statistical software	
Limited information loss	

Generalisation tends to leave the information value largely intact but reduces the precision of the information. An exact age can become an age range, location data becomes less precise and numerical variables can become labels.

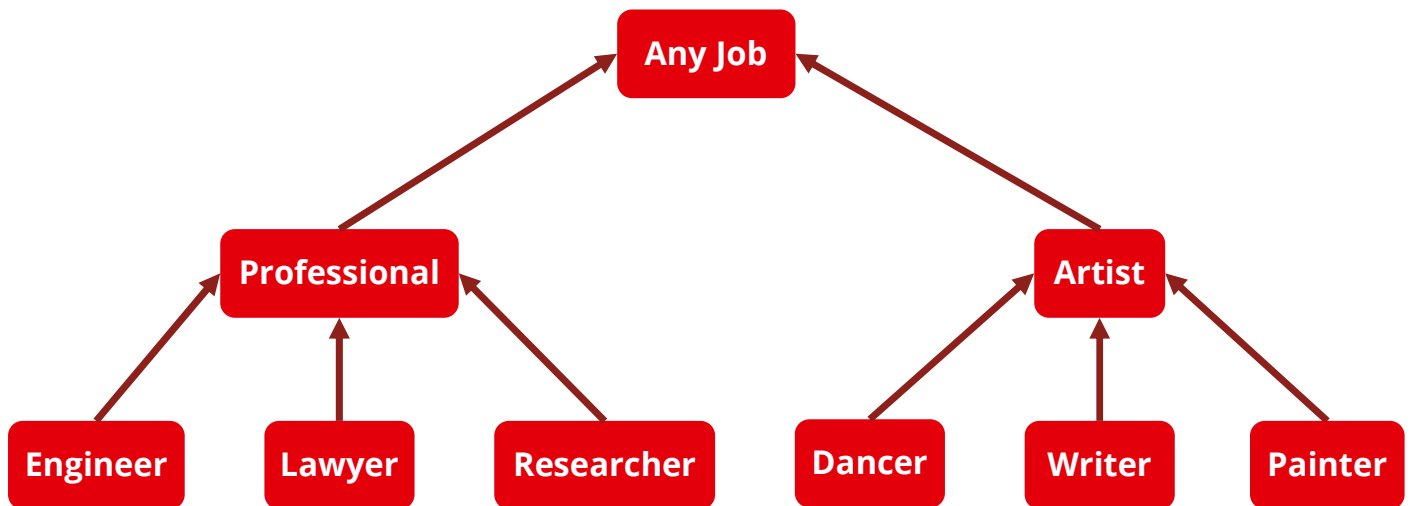
This technique grants a lot of flexibility, the scale of the generalisation can be adapted to the balance between information loss and privacy risks. This means the technique generally causes relatively little information loss. At the same time it is not always possible to do this, for example if there are only a few answer options which cannot be generalised any further (f.e. gender).

Taxonomy trees



To make the generalisation process easier you can look for existing taxonomy trees which categorise common variables in increasingly broader categories. This is one example that shows how exact ages can be classified in increasingly more generic age categories.

Taxonomy trees



This is another example of a taxonomy tree for the variable profession. It shows how the numerical example of the previous slide can be extended to other variables as well.

Top and bottom coding

Use lower and upper limits to hide outliers or extreme values

92 years  **> 65 years**

20 people  **< 100 people**

€ 238,447  **> € 100,000**

12 cats  **> 3 cats**

Benefits	Downsides
This leaves most values intact	Thresholds can be tricky to set
Works well with centered distribution	Only works with ordinal variables

Top and bottom coding is a good way to reduce the privacy risks of extreme values or outliers. It has the benefit that most values are unaffected, and works well with centered distributions. It can however be challenging to set the right thresholds for the lower or upper limits.

Sampling

Publish a representative subset of the analysed dataset

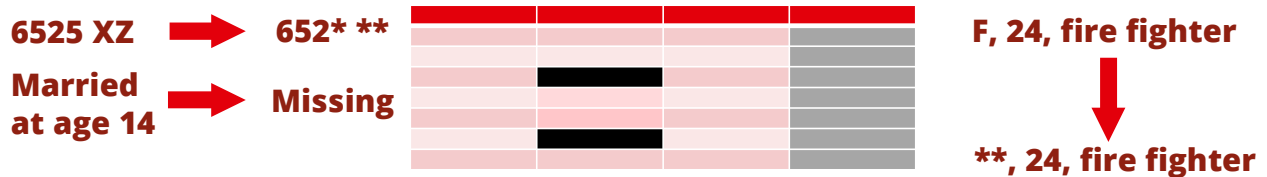
Benefits	Downsides
Adds uncertainty over inclusion	Only works with large enough samples
	Requires statistical sampling strategy
	Not helpful for extreme outliers

Sampling can be an effective strategy to increase the uncertainty whether or not a certain individual is included in the dataset. To do this you select a random sample of cases (the yellow ones in the example) which are included in the dataset.

This only works with a large sample however, while it also requires a good sampling strategy. Moreover, it does not protect against extreme outliers. If these are included in the sample people would still be identifiable.

Suppression (masking)

Suppress specific values for some units with “missing” units



Benefits	Downsides
Good way to hide outliers	Distorts frequency counts
Good way to hide rare combinations	Analysis becomes more difficult
Variable is retained in the dataset	Risk of deduction of the missing values

Suppression (or masking) is a way to to hide revealing information without deleting the variable in its entirety. This can be used to reduce the sensitivity of extreme outliers, sensitive combinations or variables which you want to make less precise. The benefit is of this technique is that the information loss is limited to what is necessary, leaving much of the data intact and the variable in the dataset. There are also risks associated with this technique, for example that it can skew frequency counts and makes analysis more difficult when too much information is masked. Lastly there is also a risk that missing values are deduced from contextual information or simply because they are the only ones missing.

Swapping

Swapping values in pairs in a fraction of the data



Benefits	Downsides
Introduces a level of uncertainty	Not suitable for all variables
Distribution and variance are maintained	Correlations between variables are lost
	Easier to decipher and reverse engineer

Swapping values can also help to introduce a level of uncertainty to the dataset. For this you take two cases which have an identical value on one variable (yellow) and swap the value of a related variable (blue). This has as major benefits that the distribution and variance of the dataset is left intact. However, it is also a technique that is not suitable for all variables, which removes the correlation between the swapped variables, and which is relatively easy to decipher.

Aggregation

Convert dataset into a record of summarised values

Donor	Income	Amount		Income	# donations	Amount
Donor 1	2200	200		1000-1999	1	400
Donor 2	1600	300		2000-2999	2	200
Donor 3	1800	100				

Benefits	Downsides
Easily done with totals or averages	Not always applicable
	Groups must be created carefully
	Often combined with suppression

Aggregation can be a useful technique with certain sets of data. To use this technique, you transform the identifiable data in more abstract aggregated data. This is useful if you only need totals or averages, but also loses some information. There is also the question of how to choose good groups in the aggregated table.

Search & replace

Replace identifiers with redactions or vague descriptors

Benefits	Downsides
Better suited for qualitative research	Error-prone if not done with care
Can be discussed with participants	Limits possibilities for re-use
Easy with “advanced search” features	

Summary: So my primary school was X which was about X minutes from my home in X. My best friends from day one were X, X and X. I found my first employment at the age of X, in the local X.

Sum<<Lucy>>: So my pri<<Lucy>> school was Oakland National School which was about 10 minutes from my home in Dunsany. My best friends from day one were <<Jenny>>, <<Lucy>> and <<Charie>>. I found my first employment at the age of 18, in the local infir<<Lucy>>.

Summary: So my primary school was @@the deathstar## which was about 10 minutes from my home in @@outer space##. My best friends from day one were @@Han Solo##, @@Chewbacca## and @@Princess Leia##. In found my first employment at the age of 18, in the local @@Jabba's Palace##

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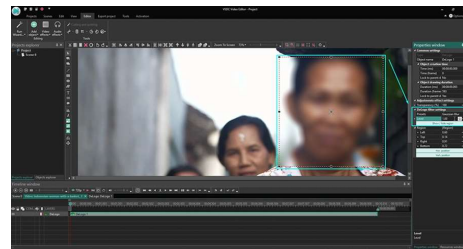
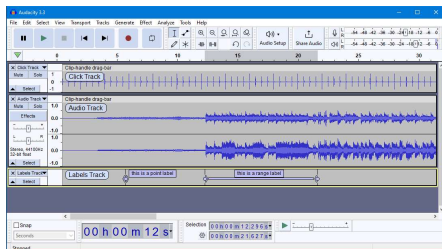
Search and replace is often the simplest technique to work with qualitative data. This can be done with readily available software and allows for a lot of flexibility in the way that research data is anonymised. There are also risks though, when search and replace is not used correctly. The loss of detailed data can also come at the expense of re-use possibilities, which can already be challenging with qualitative research data.

Three examples of what can go wrong with search and replace:

1. All identifiable data has been replaced with X's, making the text difficult to read and difficult to interpret. Furthermore, this would make it impossible to accurately document what anonymisation steps have been taking.
2. Here search and replace was used to carelessly, replacing “mary” with “Lucy”. This has replaced text in places where it should have remained intact.
3. In this example identifiable data has been replaced with Star Wars terminology, making the text unreadable and confusing.

Distortion, blurring, defacing

Transforming or distorting audio and video data



Benefits	Downsides
Only option for audio and video data	Destroys a lot of information value
	Often better to rely on access control
	Costly or time consuming

For audio and video data there are limited options for anonymisation. While there are many software solutions that can distort, transform or remove details from audio or video, these are not always suitable for research data. Often a lot of information value is lost, while the process of anonymisation can be costly, time consuming or require specialist technological skills. In many cases it is better to choose the right informed consent and access control to retain the research data without the necessity to anonymise.

TECHNOLOGY

Software solutions

Common research software



Quantitative data solutions



Qualitative data solutions



Other software options



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There are countless software solutions that can make the process of anonymisation easier. They vary immensely in functionality, complexity and accessibility, leaving it up to the researcher to find the best tools for their research project. In any case the **software that is used has to be approved for use at the Radboud University** since it is likely that the researcher will use it to process (sensitive) data which contains personal data. This is particularly true for software that is connected to the internet as this might transfer sensitive data outside of the European Union.

The simplest tools available are those that are included with the software that is already available to all researchers at the Radboud University. Word and Excel offer advanced search and replace features, while statistical software provides means to edit tables. There are however also dedicated quantitative and qualitative tools that provide more functionalities and are better suited for specifically anonymisation. For specific data types such as audio, video or MRI data there are also dedicated tools available that can make it possible to remove identifying information.

With the rise of AI and large language model technology there is also a surge in new types of software which might be suited for the process of anonymisation. It remains to be seen how these can be best used with research data both effectively and safely, but it could be a possible solution in the future.

Privacy-enhancing technologies



Federated learning



Multi-party computation



Synthetic data generation

Privacy-enhancing technologies are a growing field of interest in technological solutions that make it possible to work with sensitive identifiable data without the need to anonymise the dataset. Many of them are relatively new or still in development.

1. **Federated learning:** this is a decentralised form of machine learning which does not require a central database with sensitive data. It makes it possible to train models in a way that only shares anonymised results.
2. **Multi-Party Computation:** this functions as a shared database, where multiple partners can compute data jointly without the need to reveal data to each other. Instead the data is kept safe by a range of cryptographic techniques.
3. **Synthetic data generation:** existing data is used to generate a model that can create new simulated data. The model is based on AI, machine learning or statistical methods so that desired properties of a real life dataset are maintained.

Questions?



www.ru.nl/rdm



Data steward



rdmsupport@ubn.ru.nl

In case you have questions or need support you can first check out the RDM website of the Radboud University to see if you can find more information. For more specific questions related to your research project it is best to ask your local data steward, since they have expertise in your field, the practices that are common and the software options that can make your life easier. For questions about this presentation or the process of anonymisation in general you can reach out to the RDM Support Team via e-mail.

Datastewards (RDM)

Research Institute	Data steward	Research Institute	Data steward
Behavioural Science Institute	Rob Gommans	Radboud Institute for Biological and Environmental Sciences	Eelke Jongejans
Centre for Language Studies	Henk van den Heuvel	Radboud Institute for Culture & History	Henk van den Heuvel
Donders Institute for Brain, Cognition and Behaviour	Bernhard Englitz (DCN) Temporarily vacant (DCCN) Miriam Kos (DCC) Joanes Grandjean (DCMN)	Radboud Institute for Health Sciences	Team Datastewardship IM
Institute for Computing and Information Sciences	Gunes Acar	Radboud Institute for Molecular Life Sciences	Team Datastewardship IM (UMC) Joost Martens (FNWI)
Institute for Management Research	Francie Manhardt	Radboud Social Cultural Research	Bas Hofstra
Institute for Mathematics, Astrophysics and Particle Physics	Andrew Levan	Radboud Teachers Academy	Robin Satter
Institute for Molecules and Materials	Evan Spruijt	Research Centres of the Faculty of Law	Anita Böcker
Institute for Science in Society	Rob Lenders Riyan van den Born	Research Institute for Philosophy, Theology and Religious Studies	Sanna van Roosmalen

The datastewards are the first point of contact for researchers at the RU. They are experts on the data management practices in their own field and have the most knowledge of the local policies of their research institute. If datastewards need additional advice, they can fall back to the RDM Support team for additional support.

Please be mindful that this is a historical record of the situation at the time when the presentation was uploaded to Zenodo. Due to (future) staff changes this list could change and be no longer up-to-date.

Sources

Slide	Source
Icons	All icons are created by Eucalyp and can be found on www.flaticon.com
Deleted due to copy-right	▪ https://m.media-amazon.com/images/I/81oFt6uxG2L._SL1500_.jpg ▪ http://arthurvidich.blogspot.com/2008/11/art-vidich-and-his-candor-days.html ▪ Vidich, A. and Bensman, J. (1958) <i>Small Town in Mass Society</i>. Princeton University Press. ▪ Henry, C.A. (2000). <i>Candor Village, yesterday and today: A centennial celebration edition</i>. Self-published. p. 160. ▪ Whyte, W. (1958) Editorial. 'Freedom and responsibility in research: The 'Springdale' case'. <i>Human Organization</i>. 17 (summer), pp 1 – 2.
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